Airline Reservation System

Project Documentation · OOPROG Final Project · Group 1, Section C2B
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Overview

This is the planning and design document for our OOPROG final project: a console Airline Reservation System in Java. It covers what the system does, how it's built, who built which part, the rules every input follows, why we made each design choice, and how we tested it.

Project at a glance

Item	Detail
Course	OOPROG Finals, Section C2B
Group	Group 1: Artvincent Abesamis, Paolo Adrian Macaraig, Eam Jae Mateo, Kurt Tengonciang
System	Airline Reservation System that sells flights from four fictional airlines: Mateo Airways (MW), Air Abesamis (AB), TengoFly (TF), and PAM Aviation (PM)
Language / UI	Java 11 or newer, console only (no GUI)
Repository	final-project-1-ooprog, created by Paolo
Deadline	Repo link submitted on Canvas by Oct 23, 2026; presentation Oct 24, 2026 (Saturday) during class

Requirements and how we meet them

Requirement	How we meet it
Login	Username + password login for every role; passwords stored hashed, never as plain text
Full CRUD	Flights (Admin), user accounts (Admin), bookings (Passenger and Staff) can all be created, viewed, updated, and cancelled or removed
Multiple roles	Admin, Staff, Passenger, each with its own menu
Encapsulation	All fields private; state changes only through methods that validate first

Requirement	How we meet it
Inheritance	Passenger and Staff extend User; Admin extends Staff (User → Staff → Admin), and every checked exception extends AirlineException
Polymorphism	Each role overrides showMenu(); each fare and payment type overrides its own pricing or validation
Abstraction	Abstract User; interfaces FareStrategy and PaymentStrategy
Exception handling	Custom checked exceptions for login, flights, seats, bookings, payments, check-in, all under one parent, AirlineException
Singleton	AirlineSystem holds all data; one instance through getInstance()
Strategy	FareStrategy (Economy, Premium Economy, Business) and PaymentStrategy (Card, GCash, PayPal)
Validation	Every input re-prompts on a bad value; rules in the Input Validation section

Key decisions

- Four airlines in one booking system, each with its own planes, baggage, change fees, and pet rule. This changed from our proposal, which had a single airline.
- Up to 4 passengers per booking, all under one PNR
- Data is saved to a file, so it survives restarting the program
- Admin and Staff log in through a separate Staff Portal; passengers use the normal Login
- Console UI in the same style as our midterm project

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System Design

The program has three role menus that call feature flows, and every flow reads and writes through the single AirlineSystem, which saves to a file.

Features by role

Role	Menu options
Before login (anyone)	Login (passengers), Create Account, Flight Status, Staff Portal (Admin and Staff), Exit
Passenger	Search Flights, Book a Flight, My Bookings, Manage Booking, Check In, Cancel Booking, Change Password, Logout
Staff (Ticketing Menu)	Search Flights, Book for a Walk-in, Manage Booking, Check In, Cancel Booking, Change Password, Logout
Admin	Manage Flights (add, view, update status, reschedule, change base fare, cancel, remove), Manage Users (view, add Staff, deactivate/reactivate, unlock), Ticketing Tools (the Staff menu), All Bookings, Passenger List, Revenue Report, Change Password, Logout
Manage Booking (PNR + last name)	View e-ticket, Save e-ticket as a text file, Change seat, Change flight, Add checked baggage, Add a pet, Check in, Cancel booking

A booking holds 1 to 4 seated travelers plus infants on a lap. Each traveler’s type comes from their birthdate on the departure date. Seniors and PWDs can claim their discount, each traveler can add checked baggage, and up to 3 small dogs can fly in the cabin per flight.

Keeping Main small

Main only shows the main menu and calls user.showMenu(). Each feature lives in its own flow class (BookingFlow, FlightManagementFlow, and so on). A role menu just calls the flows it needs. That way four people rarely edit the same file, which avoids merge conflicts.

Classes

Class	Kind	What it does	Owner
Main	class	Title screen, main menu, calls showMenu()	Paolo
AirlineSystem	Singleton	Holds all users, flights, bookings; one instance through getInstance()	Paolo
SystemData, DataStore	classes	What gets saved, and saving/loading it to airline-data.ser	Paolo
User	abstract class	Username, hashed password, full name, active and locked flags; abstract getRole(), showMenu(), isEmployee()	Paolo
Passenger, Staff	subclasses of User	Passenger menu; Staff ticketing menu	Paolo
Admin	subclass of Staff	Admin menu; Ticketing Tools calls super.showMenu()	Paolo
AuthFlow	flow	Login, Staff Portal, Create Account, Change Password	Paolo
UserManagementFlow	flow	Admin: view users, add Staff, deactivate/reactivate, unlock	Paolo
BaggageOption, BaggageFlow	enum, flow	20/24/28/32 kg checked baggage, at booking or later	Paolo
ConsoleStyle	helper	Screen formatting; row() is overloaded	Paolo
InputHelper,	helper, functional	All keyboard input: re-	Paolo

Class	Kind	What it does	Owner
InputRule	interface	prompt, cancel, Y/N, numbers, dates	
InvalidInputException, OperationCancelledException	checked, unchecked exceptions	Bad input; typing cancel	Paolo
Airline	enum with fields	The 4 airlines; each has a code, a kind, its plane, free bag, biggest bag, change fee rate, and pet rule	Art
Flight	class	Airline, flight number, route, departure, status, aircraft, base fare, its own seat map, pets; toString()	Art
FlightStatus, Airport	enums	Scheduled/Delayed/ Cancelled/Departed; the 10 airports served	Art
FlightManagementFlow	flow	Admin: add, view, update status, reschedule, change base fare, cancel, remove	Art
SearchFlow, FlightStatusFlow	flows	Search by route, date, and airline; check one flight without logging in	Art
ReportFlow	flow	All bookings, passenger list, revenue report	Art
PetAddOn, PetFlow	class, flow	Small dog in the cabin, up to 10 kg, max 3 per flight	Art
AirlineException + 6 more	checked exceptions	Parent of every checked exception; one child per kind of	Art

Class	Kind	What it does	Owner
		failure	
AircraftType, CabinLayout, ColumnSlot, Seat	classes	ATR 72, A321, A321neo (all Economy), A350 seat maps: rows, letters, aisles, booked or free	Kurt
FareStrategy + EconomyFare, PremiumEconomyFare, BusinessFare	Strategy	Price, refund rule, and change fee per cabin	Kurt
FareBreakdown	class	Base fare, discount, VAT, fees, baggage, pet, total	Kurt
Booking, BookingStatus	class, enum	PNR, flight, travelers, fare, payments, status	Kurt
Traveler	class	One person: name, birthdate, type, discount, seat, baggage, pet, checked in; 2 constructors	Kurt
BookingFlow, ChangeSeatFlow, ChangeFlightFlow	flows	Book a flight; change seat; move a booking to another flight	Kurt
ETicket	class	E-ticket on screen or saved as a .txt file	Kurt
PaymentStrategy + CreditCardPayment, GCashPayment, PayPalPayment	Strategy	Checks payment details for each method	Mateo
PaymentRecord, PaymentFlow	class, flow	One payment’s receipt; the checkout screen	Mateo
PassengerType, FareDiscount	enums	Adult/Child/Infant from birthdate; Senior	Mateo

Class	Kind	What it does	Owner
		and PWD discounts	
TravelerDetailsFlow	flow	Each traveler’s name, birthdate, discount + ID	Mateo
ManageBookingFlow, CheckInFlow, CancelFlow	flows	Find booking by PNR + last name; check in + boarding pass; cancel with refund	Mateo

Checked exceptions, all extending AirlineException: InvalidLoginException, InvalidInputException, FlightNotFoundException, SeatUnavailableException, BookingNotFoundException, InvalidPaymentException, CheckInNotAllowedException. The one unchecked exception is OperationCancelledException.

Input validation

Every field’s rule, an accepted example, and the exact rejection message are in the Input Validation section. All four parts follow those rules, so bad input is rejected the same way everywhere.

Business rules

Rule	Detail
Login	Passengers use Login; Admin and Staff only the Staff Portal. An account on the wrong screen gets “Incorrect username or password.” 5 wrong passwords lock the account until an Admin unlocks it
Airlines	Mateo Airways (MW, premium full-service), Air Abesamis (AB, full-service), TengoFly (TF, low-cost), PAM Aviation (PM, regional island-hopper). A flight number starts with its airline’s code, and Admin only gets that airline’s plane when adding a flight
Booking	Only Scheduled or Delayed flights that haven’t departed
Seat lock	A seat is marked taken only after the booking is paid
Cabins	Economy x1.0, Premium Economy x1.6,

Rule	Detail
	Business x3.0 of the flight's base fare. Mateo Airways' A350: all three; Air Abesamis' A321: Business and Economy; TengoFly's A321neo and PAM Aviation's ATR 72: Economy only
Age	Worked out on the departure date, not today: a baby who turns 2 before the flight books as a Child
Children and infants	A child needs a traveler aged 18+ in the booking; no more infants than travelers aged 18+
Infant	Sits on an adult's lap: no seat, no checked baggage, Php 500 infant fee
Fees per seated traveler	12% VAT on the base fare, Php 300 fuel surcharge, Php 200 terminal fee
Senior / PWD	20% off the base fare + VAT exemption on domestic flights (RA 9994, RA 10754). Seniors must be 60+ on the departure date. One discount per person
Hand-carry	7 kg included for every seated traveler
Checked baggage	20 kg Php 600, 24 kg Php 750, 28 kg Php 900, 32 kg Php 1,050. Mateo Airways and Air Abesamis include 20 kg (bigger bags pay the difference); TengoFly charges for every bag; PAM Aviation takes bags up to 20 kg only. Can be upgraded later until 1 hour before departure (pay the difference)
Pets	Small dog up to 10 kg, Php 2,500, one per traveler aged 18+, max 3 per flight, until 1 hour before departure. PAM Aviation doesn't carry pets
Check-in	Opens 24 hours before departure, closes 1 hour before
Cancel booking	Until departure. Economy keeps Php 1,000 per seated traveler, Premium Economy Php 500, Business refunds everything
Change flight	Same airline, route, and cabin. Change fee

Rule	Detail
	per seated traveler: Economy Php 1,000, Premium Economy Php 500, Business free, then Mateo Airways charges half and TengoFly 1.5x, plus the fare difference if the new flight costs more (a lower fare isn't refunded). Not after check-in or within 1 hour of departure; refused if an age group or discount would change on the new date
Change base fare (Admin)	Only new bookings use the new fare; paid bookings keep their price
Cancel flight (Admin)	Every booking on it is cancelled with a full refund
Remove flight (Admin)	Only when it has no active bookings
Deactivate user	Can't log in; their bookings stay. An Admin can't deactivate themselves or the last active Admin
Fare shown	Base fare, discount, VAT, fees, baggage, and pet fee broken down on the review screen

Sources

- PWD and senior citizen discounts in the Philippines: 20% + VAT exemption on domestic air travel
- Cebu Pacific baggage allowance (Wego): 7 kg hand-carry, 20 kg checked + 4/8/12 kg add-ons, 32 kg max per bag, no checked allowance for infants
- PAL allows small dogs on all domestic flights (Inquirer): FurPAL rules, Feb 2026

Task Division

Each member owns 12 to 17 files and the OOP lessons inside them, and explains those in the Q&A. Nobody commits inside someone else's files.

Who owns what

Member	Files	Lessons to explain
Paolo (17 files)	Creates the repo; Main, AirlineSystem, SystemData, DataStore, User, Passenger, Staff, Admin, AuthFlow, UserManagementFlow,	Singleton, abstract class, multilevel inheritance and super.showMenu(), overloading, encapsulation, finally

Member	Files	Lessons to explain
	BaggageOption, BaggageFlow, ConsoleStyle, InputHelper, InputRule, InvalidInputException, OperationCancelledException	
Art (17 files)	Airline, Flight, FlightStatus, Airport, FlightManagementFlow, SearchFlow, FlightStatusFlow, ReportFlow, PetAddOn, PetFlow, AirlineException and its 6 other exceptions	Exception hierarchy, throw / throws, toString(), enums (including an enum with fields: Airline), overloading, CRUD
Kurt (16 files)	ColumnSlot, CabinLayout, AircraftType, Seat, FareStrategy, EconomyFare, PremiumEconomyFare, BusinessFare, FareBreakdown, BookingStatus, Traveler, Booking, BookingFlow, ChangeSeatFlow, ChangeFlightFlow, ETicket	Strategy (fares), interface default and static methods, constructor overloading with this(...), writing a file
Mateo (12 files)	PaymentStrategy, CreditCardPayment, GCashPayment, PayPalPayment, PaymentRecord, PaymentFlow, PassengerType, FareDiscount, TravelerDetailsFlow, ManageBookingFlow, CheckInFlow, CancelFlow	Strategy (payments), interfaces, multi-catch, dates and ages

Work order

1. **Setup (Sun Oct 18).** Paolo creates final-project-1-ooprog and adds the other 3 as collaborators. Everyone checks that their git email matches GitHub.
2. **Foundations (Sun Oct 18 to Mon Oct 19).** First files: Paolo’s input helpers and ConsoleStyle, Art’s exceptions and enums, Kurt’s seat and cabin classes, Mateo’s

PassengerType, FareDiscount, and PaymentStrategy. These compile on their own, so all 4 can work at once.

3. **Features (Mon Oct 19 to Wed Oct 21).** The model classes, then the flows and menu options on top of them.
4. **Integration and testing (Thu Oct 22).** All four parts are pushed, then the whole group runs the Group Test Guide on the final program. Fix bugs in your own files.
5. **Submit and study (Fri Oct 23).** Submit the repo link on Canvas. Each member practices explaining their own classes.
6. **Presentation (Sat Oct 24).** During class.

How We Used GitHub

Repository: final-project-1-ooprog, created by Paolo. All four members are collaborators and commit under their own GitHub accounts.

```
final-project-1-ooprog/
├── README.md
├── .gitignore
├── TEST-ACCOUNTS.txt
├── Macaraig/              Paolo's 17 files
├── Abesamis/              Art's 17 files
├── Tengonciang/           Kurt's 16 files
├── Mateo/                 Mateo's 12 files
└── documentation/         this document, the sample run, and the diagrams
```

Our rules:

- Each member commits only their own files, in their own surname folder.
- git pull before starting work and again before pushing, so nobody overwrites someone else's changes.
- Add files by name (git add Tengonciang/Seat.java), never git add ., so compiled .class files and other people's work never get committed by accident.
- Compile before pushing.
- One logical change per commit, with a message that starts with a verb: Add Seat class, Fix seat staying booked after cancel.
- No file has a package line, so the four folders compile together as one program: javac -d out Macaraig/*.java Abesamis/*.java Tengonciang/*.java Mateo/*.java, then java -cp out Main.
- The .gitignore keeps out/, .class files, the save file airline-data.ser, and saved e-tickets out of the repo.

Conventions and Testing

Four people writing one program need to make it look like one person wrote it: same folder layout, same input behaviour, same screen style.

Code rules

- One public class per file; file name = class name.

- All fields private; getters only where another class needs them.
- Only InputHelper reads the keyboard. No new Scanner(System.in) anywhere else.
- Model classes (Flight, Booking, Seat, …) never print. Flows print.
- Model classes throw our custom exceptions; flows catch them and show the message.
- No magic numbers: limits and fees are named constants (MAX_TRAVELERS = 4).
- Comments short and plain, explaining *why*.

Screen style

Every screen starts with a header bar. The left side is a breadcrumb of where you are; the right side shows who is logged in, or today's date when nobody is. All of it comes from ConsoleStyle, so nobody prints their own headers.

```
======================================================================
 Passenger > Manage Booking                              Andrea Cruz
----------------------------------------------------------------------

  (Type 'cancel' at any prompt to go back)

PNR: ZWZAMR
Last name of the first traveler: Garcia
[!] No booking found with that PNR and last name.

PNR: ZWZAMR
Last name of the first traveler: Cruz

  [ Booking ZWZAMR - Confirmed ]
  MW701   Manila (MNL) -> Cebu (CEB)   2026-10-19 18:30   Premium Economy

    Andrea Cruz                 Adult   11A   not checked in
    Jose Cruz                   Adult   11C   not checked in

  Paid: Php 15,209.20

  1 - View e-ticket
  ...
  9 - Back

Choice: 3

-- Change Seat --

  1 - Andrea Cruz (seat 11A)
  2 - Jose Cruz (seat 11C)

Choice: 2
```

- Screen headers: ConsoleStyle.header("Title"). Menus call ConsoleStyle.enter("Name") when they open and leave() in a finally when they close, which is how the breadcrumb builds itself.
- Smaller headings inside a screen: ConsoleStyle.section("Name"), which prints -- Name -- with a blank line above and below.
- Error lines start with [!]: always print them as ConsoleStyle.error(message).

- Spacing: one blank line under every header, menu options indented 2 spaces, a blank line before Choice:, and a blank line after an error before the prompt asks again. InputHelper handles the last two.
- Use InputHelper.confirm(...) for the last Y/N before something happens (pay, cancel, create), so the result message gets its own line. Use readYesNo(...) for other Y/N questions.
- Tables, seat maps, e-tickets and boarding passes keep their own fixed layout.
- Every multi-step flow accepts cancel at any prompt (ConsoleStyle.cancelHint() shows the reminder).
- Bad input shows the reason and asks again. It never returns to the menu or crashes.
- Anything that changes data shows a review screen and asks (Y/N) first.
- Money: Php 1,234.00. Dates: YYYY-MM-DD. Times: HH:MM (24-hour).
- Messages are short and specific: Seat 12A is already taken., not Error!.

Before pushing

- The whole program still compiles from the repo's main folder
- You ran your own feature: the normal path, every bad input, and cancel
- Only your own files are in the commit

Test tracker

Each feature is tested by someone who didn't build it.

Feature	Built by	Tested by
Login, Staff Portal, Create Account, Change Password	Paolo	Art
User management	Paolo	Mateo
Saving and loading data	Paolo	Kurt
Checked baggage	Paolo	Mateo
Add, update, reschedule flight, change base fare	Art	Kurt
Airlines: codes, search filter, free bags, change fee, no pets on PAM	Art	Kurt
Cancel and remove flight	Art	Mateo
Search flights and Flight Status	Art	Paolo
Reports	Art	Paolo
Pets in cabin	Art	Kurt

Feature	Built by	Tested by
Seat maps	Kurt	Art
Booking with 1-4 travelers + infants	Kurt	Mateo
Change seat	Kurt	Paolo
Change flight	Kurt	Art
E-ticket on screen and saved as .txt	Kurt	Mateo
Passenger types and Senior/PWD discount	Mateo	Kurt
Payments	Mateo	Kurt
Manage booking, cancel, refund	Mateo	Art
Check-in and boarding pass	Mateo	Paolo

Input Validation

Every input is checked where it’s typed, rejected with a specific reason, and asked again. The program never crashes and never kicks the user back to the menu over a typo.

Rules that apply to every prompt

- Leading and trailing spaces are trimmed; repeated spaces inside names collapse to one.
- cancel at any prompt backs out to the menu without saving anything.
- Numbers are checked with a pattern before parsing, so NaN, 1e3, 0x10 and Infinity never get through.
- Dates must be real calendar dates: 2026-02-30 is rejected, not rolled over to March.
- Uniqueness (username, flight number) is checked right at the prompt, not after five more questions.
- The model classes check the same rules again in their constructors, so bad data can’t get in through any other path.
- If input ends unexpectedly (Ctrl+D / Ctrl+Z), the program saves and exits cleanly instead of crashing.

Accounts

Field	Rule	Accepted	Rejected → message
Username	5-15 characters, letters, digits, _; starts with a letter; unique, case doesn't matter	kurt_t	kt → Username must be 5-15 characters. Admin (exists) → That username is taken.
Password	8-20 characters, at least 1 letter and 1 number, no spaces; typed twice	fly2Cebu	password → Password needs at least one number. Mismatch → Passwords don't match.
Full name	Letters (incl. ñ, é), spaces, . - '; starts with a letter; 2-50 characters	Ma. Clara Niño-Reyes	123, !!! → Name can only have letters, spaces, periods, hyphens, and apostrophes.
Login and Staff Portal	Same message whether the username, the password, or the screen is wrong (passengers use Login, Admin and Staff the Staff Portal)		Incorrect username or password.
Change Password	Current password must be right; the new one follows the Password rule and must be different	fly2Davao	Wrong current → Current password is incorrect. Same as before → The new password must be different from the current one.

Flights (Admin)

Field	Rule	Accepted	Rejected → message
Flight number	The chosen airline's	PM777 (for PAM	MW12 for PAM

Field	Rule	Accepted	Rejected → message
	code (MW, AB, TF, PM) + 1-4 digits, stored uppercase, unique	Aviation)	Aviation → PAM Aviation flight numbers are PM and 1-4 digits, e.g. PM123. PM301 (exists) → Flight PM301 already exists.
Origin / destination	Picked from the airport list, not typed; destination differs from origin	2 - Cebu (CEB)	Same as origin → Destination can't be the same as the origin.
Departure date	Real date, not in the past, at most 1 year ahead	2026-11-20	2026-13-01 → Not a real date. Past → The date can't be in the past.
Departure time	HH:MM, 24-hour; date + time must be in the future	08:30	8:30, 25:00 → Use 24-hour HH:MM, e.g. 08:30.
Base fare	Above 0, at most 100,000, up to 2 decimals	2499.50	0, -5, 3500.555 → Fare must be above 0 with at most 2 decimals.
Reschedule	Flight not departed or cancelled; new date/time in the future		Departed → This flight has already departed.

Change base fare uses the same Base fare rule. Flight Status accepts any airline's flight number: AR999 → Flight number must be an airline code (MW, AB, TF, PM) and 1-4 digits, e.g. MW101. MW999 → Flight MW999 not found.

Travelers

Field	Rule	Accepted	Rejected → message
Seated travelers	1-4 per booking	2	5 → A booking holds 1 to 4 seated travelers.

Field	Rule	Accepted	Rejected → message
Birthdate	Real date, not in the future, at most 120 years ago	2015-06-30	Future → Birthdate can’t be in the future.
Infants	At most 1 per traveler aged 18+		2 infants, 1 adult → Each infant needs their own adult (18+) in the booking.
Children	A booking with a child or infant has at least one traveler aged 18+		Children only → Children and infants must travel with an adult (18+).
Senior discount	Traveler is 60+ on the departure date		Age 59 → The senior discount is for travelers 60 and older on the departure date.
Senior / PWD ID	Needed when claiming a discount: 4-20 letters, digits, -	PWD-123456	Blank → Enter the ID number to claim this discount.
Seat	Exists on this aircraft, in the chosen cabin, free, not picked twice in one booking	12A	13A → Seat 13A doesn’t exist on this aircraft. 1A in Economy → Seat 1A is in Business, not Economy.
Checked baggage	Menu choice; infants have no allowance	2 - 20 kg	Infant → skipped automatically
Pet weight (stretch)	Above 0, up to 10 kg, up to 2 decimals; max 3 pets per flight, 1 per passenger	6.5	12 → Only dogs up to 10 kg can fly in the cabin. 4th pet → This flight already has 3 pets.

Payment

Field	Rule	Accepted	Rejected → message
Card number	16 digits, spaces or dashes allowed and removed; passes the Luhn check; shown as **** 1111	4111 1111 1111 1111	1234567812345678 → That card number isn't valid.
Expiry	MM/YY, month 01-12, not expired	08/28	13/27 → Use MM/YY. 01/24 → This card has expired.
CVV	3 digits; never saved or printed	123	12a → CVV must be 3 digits.
GCash	09 + 9 digits, spaces allowed	0917 123 4567	0817... → GCash numbers start with 09 and have 11 digits.
PayPal	Valid email format	kurt@gmail.com	kurt@gmail → Enter a valid email address.

Bookings

Field	Rule	Accepted	Rejected → message
PNR	6 characters, no I or O; case doesn't matter	x7b9tq	ABC → PNR must be 6 letters or numbers.
PNR + last name	Must match the lead traveler; same message whether the PNR or the name is wrong		No booking found with that PNR and last name.
Check-in	Booking active, 24 h to 1 h before departure		Too early → Check-in opens 24 hours before departure.
Cancel	Booking active, flight not departed		Departed → This flight has already departed and can't be cancelled.

Field	Rule	Accepted	Rejected → message
Y/N	Y, N, yes, no, any case	y	maybe → Please enter Y or N.
Menu choice	Whole number within the menu	2	0, 9, abc → Invalid choice. Enter a number from 1 to 6.

Change Flight: someone checked in → Flight changes after check-in are done at the airport counter. Within 1 hour of departure → Flight changes close 1 hour before departure. An age group or discount would change → the reason, then Please cancel this booking and book the new date instead.

Design Rationale

Every feature is here either because real airlines work that way or because one of our past projects broke without it. This section answers “why did you do it like this?” for each one.

Scope and assumptions

Stated up front, so the professor knows what’s out of scope by choice, not by accident:

- Domestic Philippine flights only, one time zone (Philippine Time), prices in PHP
- One-way bookings; a return trip is a second booking
- Payments are simulated: details are validated, but no money moves and card numbers are never saved
- Traveler names can’t be changed after booking, same as real airlines; cancel and rebook instead

Why each feature exists

Feature	Why	Real-world basis
6-character PNR without I or O	Short, unique, and never confused with 1 or 0 when read aloud	Airline booking codes
Manage Booking by PNR + last name	No login needed, and the same error for a wrong PNR or name doesn’t reveal which bookings exist	Every airline’s Manage Booking page
Seat maps per aircraft type, no row 13, no letter I	An ATR 72 and an A350 really are laid out differently	IATA seating conventions
Seat locked only on confirm	Abandoning a booking	Fix from our airline prototype

Feature	Why	Real-world basis
	halfway never leaves a seat stuck as taken	
Up to 4 travelers + infants on lap	Families book together; infants don’t take a seat	Standard airline infant rules
Age worked out on departure date	A traveler’s type is what they are on the flight, not on the booking day	Airline fare rules
Senior / PWD 20% + VAT exemption	Required by law on domestic flights	RA 9994, RA 10754
Checked baggage tiers up to 32 kg	Mirrors how local carriers sell baggage	Cebu Pacific baggage add-ons
Check-in 24 h to 1 h before	Too early or too late isn’t allowed at real airports	Common airline check-in window
Refund rules per fare class	Cheaper fares have stricter cancellation, as with real fare families	Airline fare families
Airports picked from a list	Typed cities break search (Cebu vs cebu city vs Cebuu)	Airline booking forms
Fare breakdown before paying	The customer sees exactly what they pay for	All-in fare display
Deactivate instead of delete	Keeps the history of bookings a user made	Account handling in real systems
Hashed passwords	Plain-text passwords are a basic security failure	Standard practice
Saved to a file	A real system doesn’t lose every booking when it restarts	Persistence
Strategy for fares and payments	A new fare class or payment method is a new class, not another if/else	GoF Strategy
One flow class per feature	Four people rarely edit the same file, so fewer merge conflicts	Team workflow

Added for the finals:

Feature	Why	Real-world basis
Staff Portal separate from Login	Employees and customers never share a login screen; an account on the wrong screen is refused without saying it exists	Airline staff systems are separate from the customer website
Admin extends Staff	An Admin can do everything Staff can, so it reuses the Staff menu with super.showMenu() instead of copying it	Managers can also work the ticketing counter
Flight Status without logging in	People check a flight before they have an account	Every airline's Flight Status page
Change Flight with fee + fare difference	Rebooking to another date is one of the most common requests; a lower fare isn't refunded	Cebu Pacific and PAL rebooking rules
Change Flight refused if an age group changes	A baby who turns 2 on the new date needs a seat and a new fare	Fare rules are based on age on the travel date
Change base fare affects new bookings only	Paid bookings keep the price they paid	Airline fare updates
Save e-ticket as a .txt file	The customer keeps a copy outside the program	Emailed itinerary receipts
Change Password	Every account system lets users change their own password	Standard practice
Four airlines in one system	An airport and a booking site sell more than one airline. Each airline has its own planes, free bag, change fee, and pet rule, so the same route can cost and work differently. A ticket can only move to the same airline	Full-service (PAL), low-cost (Cebu Pacific), and regional (Cebgo, AirSWIFT) carriers in the Philippines
Header bar with breadcrumb,	You always see where you	Breadcrumbs on websites

Feature	Why	Real-world basis
[!] errors, spaced-out menus	are and who is logged in; errors stand out from normal text; one ConsoleStyle class makes every member's screens look the same	and kiosk screens

Lessons built into the design

Lesson	Rule in this project
Sending the user back to the menu over one typo is frustrating	Every prompt retries until the input is valid or the user types cancel
A seat held before payment stays stuck if the user quits halfway	Seats are locked only when the booking is confirmed
A seat map printed from a plain list loses its aisle gaps	The map is drawn from the cabin layout template
Very large numbers can overflow calculations	Every number field has an upper limit
Commits made with a mismatched git email don't count for their author	Everyone checks their git email on day 1
Separate versions built without a shared plan are hard to merge	Task division before anyone codes
A program tied to one editor can fail on the grader's machine	It runs from any terminal
Java's built-in exceptions don't say what went wrong in airline terms	Custom checked exceptions for each kind of failure

What we added and what we left out

Addition	Why it helps	Effort	Owner	Call
README.md with diagrams and test accounts	Tells anyone how to run it	Low	All	Done
Sample data and TEST-ACCOUNTS.txt	The demo never starts from an empty screen	Low	Paolo	Done
E-ticket, also	Like a real	Low	Kurt	Done

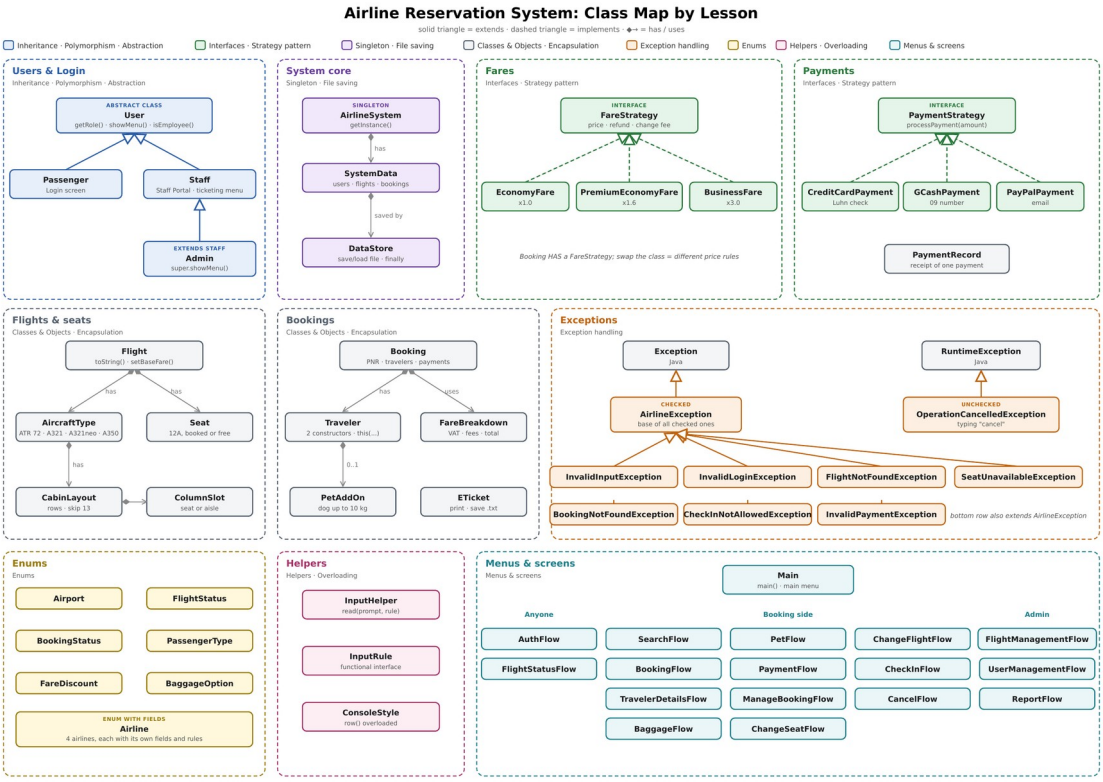
Addition	Why it helps	Effort	Owner	Call
saved as .txt	confirmation			
Account lock after 5 wrong passwords (Admin unlocks)	Real login security	Low	Paolo	Done
Flights marked Departed once their time passes	Status always matches the clock	Low	Art	Done
Flight Status, Change Password, Change Flight, Change base fare	Realistic features that each show an OOP lesson	Low to medium	Art, Paolo, Kurt	Done
Hidden password typing	Looks real, but doesn't work in the VS Code debug console	Low	Paolo	Skip
Temporary password for new Staff accounts	Realistic, but adds a whole flow to explain this week	Medium	Paolo	Skip
Round trips and multi-city	Doubles the booking logic	High		Skip
Loyalty miles, promo codes	Little rubric value	Medium		Skip
International flights and passports	Discount laws and rules change completely	High		Skip

Diagrams

The diagram files are also in the repo, in documentation/diagrams.

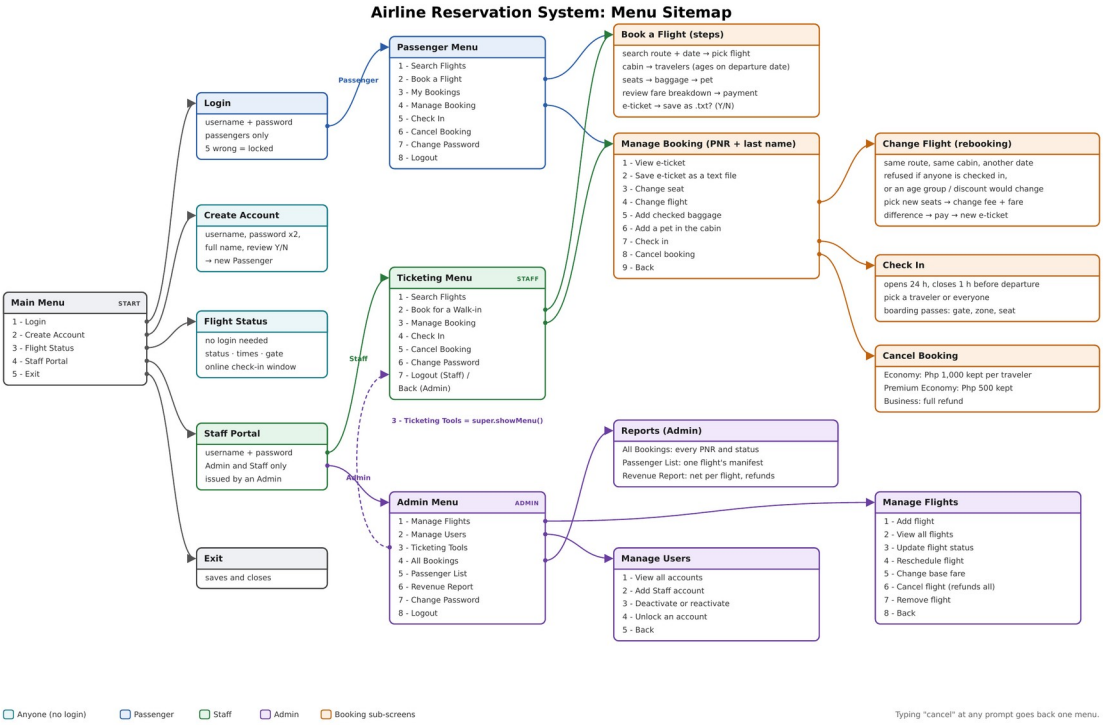
Class map by lesson

Every class, grouped by owner and colored by the OOP lesson it shows.



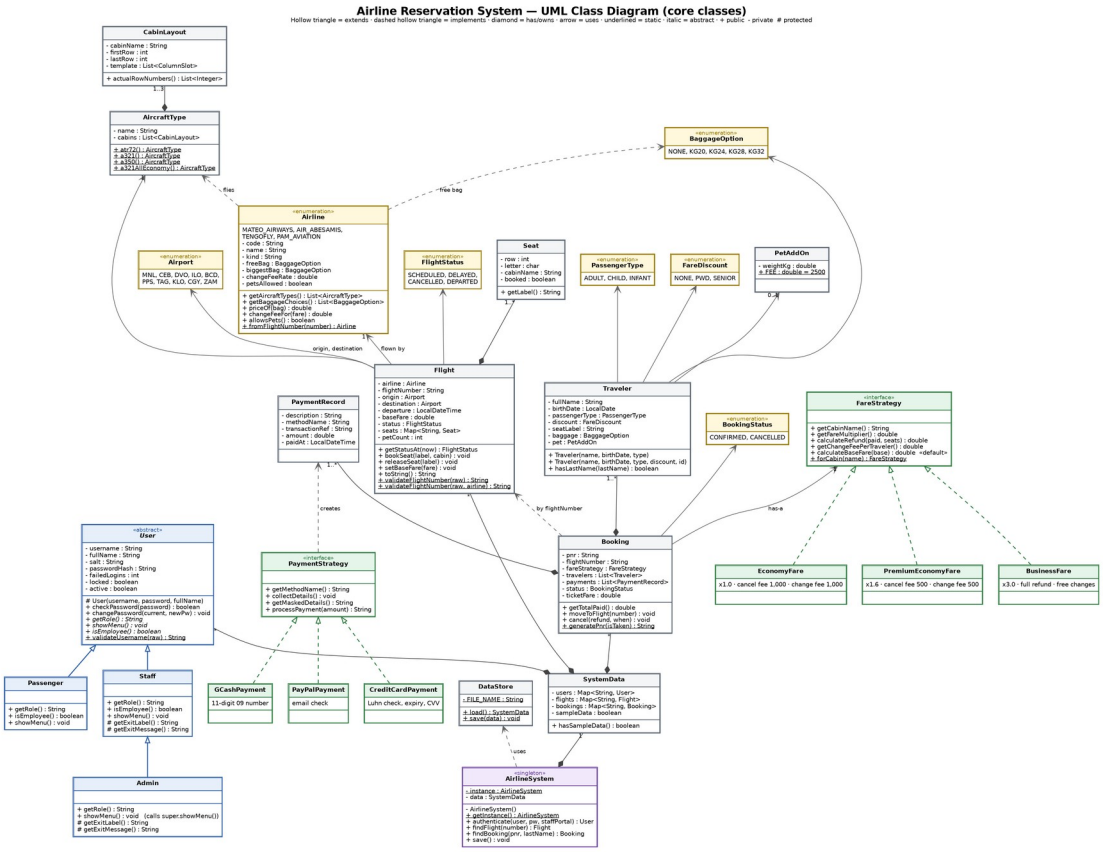
Menu sitemap

Every screen in the program and where each option leads. Admin’s Ticketing Tools opens the same ticketing menu Staff uses, through super.showMenu().



UML class diagram

The core classes with their main fields and methods. + public, - private, # protected; underlined = static; italic = abstract.



Group Test Guide

We ran this on Thursday, Oct 22, on the finished program, before submitting. Each feature is tested by someone who didn't build it (see the test assignments in Conventions and Testing). When something didn't match, the tester reported the section and step number, and the owner fixed it in their own file.

Setup (everyone)

1. git pull to get the latest version.
2. Delete airline-data.ser if it's there, so you start with fresh sample data.
3. From the repo's main folder: javac -d out Macaraig/*.java Abesamis/*.java Tengonciang/*.java Mateo/*.java, then java -cp out Main (or open Macaraig/Main.java in VS Code and click Run).
4. You should see the title screen and "Sample data loaded. Test accounts are listed in TEST-ACCOUNTS.txt."
5. Keep TEST-ACCOUNTS.txt and the sample run open. The sample run shows what every screen should look like.

Dates depend on the day you run it. "Day +2" below means two days after today. AB101 leaves about 6 hours after the first run, so do section 7 (check-in) on the same run.

1. Main menu and Flight Status

#	Do this	You should see
1.1	Main menu: type 7, then abc	[!] Invalid choice. Enter a number from 1 to 5. both times
1.2	3 Flight Status, type xyz	[!] Flight number must be an airline code (MW, AB, TF, PM) and 1-4 digits, e.g. MW101.
1.3	Type MW999	[!] Flight MW999 not found.
1.4	Type MW205	Mateo Airways, Status Delayed, check-in opens time
1.5	3 again, AB901	Air Abesamis, Cancelled, “refunded in full”
1.6	3 again, ab101 (lowercase)	Scheduled, “Online check-in Open until …”
1.7	3 again, type cancel	Back to the main menu

2. Create Account and Login

#	Do this	You should see
2.1	2 Create Account, username ab	[!] Username must be 5-15 characters.
2.2	Username admin	[!] That username is taken.
2.3	Username tester01, password short	[!] Password must be 8-20 characters.
2.4	Password password	[!] Password needs at least one number.
2.5	Password test1234, confirm test12345	[!] Passwords don't match. Try again.
2.6	test1234 twice, full name 123	[!] Name can only have letters, ...
2.7	Full name Ana Santos, answer N	“Account not created.”
2.8	Do it again and answer Y	“Account created. You can

#	Do this	You should see
		now log in.”
2.9	1 Login as admin / admin123	[!] Incorrect username or password. (Admin must use the Staff Portal)
2.10	Login as tester01 / test1234	“Welcome, Ana Santos (Passenger)”, header shows “Passenger” and her name

3. Search

Logged in as tester01.

#	Do this	You should see
3.1	3 My Bookings	“You have no bookings yet.”
3.2	1 Search: Manila → Cebu, date day +4, Any airline	“No flights …” and a “Same route within 3 days” list with airline names
3.3	Search Manila → Cebu, day +2, Air Abesamis	Only AB105, and “Air Abesamis: 20 kg checked bag included, pets allowed.”
3.4	Search Manila → Cebu, day +2, Any airline	TF111, AB105, MW701, TF115 with an Airline column and one line per airline
3.5	Search with a date in the past	[!] The date can't be in the past.

4. Booking (Mateo Airways, Premium Economy)

#	Do this	You should see
4.1	2 Book: Manila → Cebu, day +2, Any, pick MW701	Cabin list with Business, Premium Economy, Economy
4.2	Premium Economy, travelers 5	[!] A booking holds 1 to 4 seated travelers.
4.3	2 travelers: an adult, then someone born 1960	Senior Citizen appears as choice 3 only for the second one

#	Do this	You should see
4.4	Pick Senior, ID SC-12345; 1 infant born 8 months ago	Infant name and birthdate asked
4.5	Seat 13A, then 30A, then 1A	“doesn’t exist”, “is in Economy”, “is in Business”
4.6	11A for both travelers	Second one: “already picked for another traveler in this booking”
4.7	Baggage	“Mateo Airways includes a 20 kg checked bag”; 20 kg shows “Included”, 24 kg Php 150.00
4.8	Pet: Y, weight 12, then 6.5	[!] Only dogs up to 10 kg can fly in the cabin., then “Pet added”
4.9	Review	Airline, flight, fare breakdown; Senior has 20% off and “VAT (exempt)”; infant fee Php 500.00
4.10	Card 4111 1111 1111 1112	[!] That card number isn't valid.
4.11	Card 4111 1111 1111 1111, expiry 01/24, then 12/28, CVV 12, then 123, Y	“This card has expired”, “CVV must be 3 digits”, then “Payment approved”
4.12	After confirmation	“Booking confirmed! Your PNR is …” and the e-ticket with “Mateo Airways” in the header
4.13	Save the e-ticket: Y	“E-ticket saved as e-ticket-PNR.txt in the program’s folder.” The file exists.

Write down the PNR.

5. Manage Booking

#	Do this	You should see
5.1	4 Manage Booking, your PNR, last name Garcia	[!] No booking found with that PNR and last name.
5.2	Your PNR and the first traveler's last name	The booking card and options 1-9
5.3	3 Change seat, pick the second traveler, seat of the first one	[!] Seat ... is already taken.
5.4	A free seat, Y	"Seat changed to …" and a fresh header with the updated card
5.5	4 Change flight	"No other Mateo Airways flights on this route …"
5.6	5 Add baggage, upgrade the 20 kg bag to 24 kg, pay with GCash 0917123456, then 09171234567	Php 150.00 due, "GCash numbers start with 09 and have 11 digits", then approved
5.7	7 Check in	[!] Check-in opens 24 hours before departure, at ...
5.8	9 Back	Passenger menu

6. Cancel and Change Password

#	Do this	You should see
6.1	6 Cancel Booking, your PNR	Total paid, cancellation fee Php 1,000.00 (2 x Php 500), refund, the Premium Economy rule
6.2	Y	"Booking … cancelled. Php … will be refunded …"
6.3	6 again on the same PNR	[!] This booking is already cancelled.
6.4	7 Change Password, current wrong123	[!] Current password is incorrect.
6.5	Current test1234, new	[!] The new password must

#	Do this	You should see
	test1234	be different ...
6.6	Current test1234, new newpass99	“Password changed.”
6.7	8 Logout	Main menu, header back to “Home” with the date

7. Check In (same run as setup)

#	Do this	You should see
7.1	Log in as passenger01 / passenger123, 5 Check In, RYS4K7 / Reyes	Miguel and Bea listed, plus “Everyone above”
7.2	Everyone above, Y	“Check-in complete.” and boarding passes; Miguel’s shows “With infant” Lia; header says AIR ABESAMIS
7.3	5 again, same PNR	“Everyone in this booking is already checked in.” and an offer to print again
7.4	4 Manage Booking, RYS4K7 / Reyes, 4 Change flight	[!] Someone in this booking is already checked in ...
7.5	Logout	

8. Lockout

#	Do this	You should see
8.1	1 Login, tester01 with 5 wrong passwords	The 5th: “Too many wrong passwords. This account is now locked.”
8.2	Right password	“This account is locked … Ask an Admin to unlock it.”
8.3	Type cancel	Main menu

9. Staff Portal and walk-in booking (TengoFly)

#	Do this	You should see
9.1	4 Staff Portal, passenger01 / passenger123	[!] Incorrect username or password.
9.2	staff01 / staff123	“Welcome, Carla Mendoza (Staff)”, Ticketing Menu 1-7
9.3	2 Book for a Walk-in, Manila → Cebu, day +2, Any, TF111	“TengoFly flies Economy only on this plane.” (no cabin question)
9.4	1 traveler, no infants, any seat	Seat map rows 1-38, no row 13
9.5	Baggage	“TengoFly doesn’t include a checked bag”; “No checked bag” is choice 1
9.6	PayPal raf@mail, then a real-looking email	[!] Enter a valid email address., then approved

Write down this PNR too.

10. Change Flight

Still as staff01.

#	Do this	You should see
10.1	3 Manage Booking, the TengoFly PNR, 4 Change flight	Only TF115 is offered (same airline, same route)
10.2	Pick TF115, any seat	Change fee Php 1,500.00 (TengoFly Economy) + fare difference Php 336.00
10.3	Pay with GCash	“Flight changed! Your PNR stays …”
10.4	3 Manage Booking, GRC8M2 / Garcia, 4 Change flight, MW203	Change fee Php 0.00 (Business), “the new fare is lower; the difference isn’t refunded”, no payment asked
10.5	9, then 7 Logout	

11. Admin: flights

#	Do this	You should see
11.1	4 Staff Portal, admin / admin123	Admin Menu 1-8, header “Admin” and “System Administrator”
11.2	1 Manage Flights, 1 Add flight, airline PAM Aviation, number AB777	[!] PAM Aviation flight numbers are PM and 1-4 digits, e.g. PM123.
11.3	PM301	[!] Flight PM301 already exists.
11.4	PM777, Manila → Bohol, a past date, then day +5, time 25:00, then 09:30	“The date can’t be in the past”, “Use 24-hour HH:MM”
11.5	Flight time 20, then 75	“must be from 30 to 300”; then “Aircraft: ATR 72-600 … (PAM Aviation only flies this plane)”
11.6	Fare 0, then 2750, Y	“Base fare must be above 0.”, then “Flight PM777 added.”
11.7	2 View all	PM777 in the list
11.8	3 Update status, PM305, Delayed	“Flight PM305 is now Delayed.”
11.9	4 Reschedule PM401 to day +5 11:15	Rescheduled, status Delayed
11.10	5 Change base fare TF111 to 3150	“Existing bookings keep the price they paid”, then changed
11.11	6 Cancel flight TF501	Flight cancelled, 0 bookings refunded
11.12	7 Remove flight AB101	[!] ... has 1 active booking(s). Cancel the flight first ...
11.13	7 Remove flight TF501, Y	“Flight TF501 removed.”

12. Admin: users, tools, reports

#	Do this	You should see
12.1	2 Manage Users, 1 View all	Every account with role and status; tester01 is Locked
12.2	2 Add Staff staff02 / staff2pass / Nina Ramos	“They log in through the Staff Portal.”
12.3	3 Deactivate admin	[!] You can't deactivate your own account.
12.4	3 Deactivate passenger01, then reactivate	“is now deactivated”, then “is now active”
12.5	4 Unlock tester01	“tester01 is unlocked.”
12.6	5 Back, 3 Ticketing Tools	Header “Admin > Ticketing Tools”, last option “Back to Admin Menu”
12.7	7 Back, 4 All Bookings	Every PNR including the cancelled one
12.8	5 Passenger List, AB101	Miguel, Bea (and Lia on lap), checked in Yes
12.9	6 Revenue Report	Per flight, a total for each of the 4 airlines, refunds, net revenue
12.10	8 Logout, 5 Exit	“Thank you for flying with us. Goodbye!”

13. Saving

#	Do this	You should see
13.1	Run the program again (don’t delete the save file)	No “couldn’t be read” error; log in as tester01 with newpass99 works
13.2	Admin → Manage Flights → View all	PM777 is still there, TF501 is still gone

Sample Run

A full session of the finished program, from the title screen to Exit, showing every role, every main feature, and the error messages for bad input. It’s in the repo as documentation/Airline_Sample_Run.pdf (33 pages).